

Original PYQ Style Practice Bank

Common Preparation / DSA & Placement Prep

Study Hub Premium Placement Pack

This is original study material for preparation. It is not a copied official or proprietary company paper.

Premium Study System

- Track: Placement Common Preparation.
- Pack type: PYQ / Practice.
- Target outcome: selection-ready confidence for Graduate Engineer / Analyst.
- Core preparation areas: aptitude, reasoning, verbal, DSA, CS fundamentals, resume, project interview, mock tests, communication, HR stories.
- Use this pack like a mini-book: concept, table, example, practice, analysis, revision.
- Every topic should produce proof: solved questions, mock score, mistake note, interview answer, or resume evidence.

Output Quality Rules

- Read the concept in your own words, then solve without looking at notes.
- Convert every important idea into a comparison table, checklist, or one-line rule.
- Mark important traps as Common Mistake before the mock, not after losing marks.
- Keep a PYQ-style log: question pattern, topic, time taken, mistake reason, retest date.
- Use simple English; add Hindi/Hinglish meaning in brackets when it helps recall.

Priority Matrix

Area	Premium Standard	Proof Before Moving On
Aptitude	Formula, shortcut, accuracy, time control	25 mixed questions with 70%+ accuracy
Programming Logic	Loops, arrays, strings, SQL basics	Trace output and write simple code cleanly
Project	Problem, architecture, your contribution, tradeoff	2-minute story plus deep-dive answers
Communication	Crisp intro, structured HR stories, calm tone	5 recorded answers reviewed once

Exam Tip

- Service assessments reward speed with accuracy. Attempt high-confidence sections first and protect time for easy marks.

Common Mistake

- Spending too long on one aptitude question reduces total score. Set a strict skip rule and revisit only after easy wins.

Original PYQ Style Practice

- This set is designed for service based companies style campus preparation: aptitude speed, programming logic, SQL, CS basics, communication, and HR clarity.

Section A: Quantitative Aptitude

- Q1. A candidate solves 60% of a set in 36 minutes. Estimate the total time for the full set and explain the shortcut.
- Q2. A, B, and C invest in ratio 3:4:5. After 8 months A doubles investment. Find final profit ratio after 12 months.
- Q3. A pipe fills a tank in 12 hours and another empties it in 18 hours. If both are open, find

net time and the faster shortcut.

- Q4. In a class, average marks rise from 64 to 67 when one student scoring 28 is replaced. Find the new student score for class strength 40.

Section B: Reasoning and Verbal

- Q5. Five candidates sit in a row. A is left of B, C is not at an end, and D is right of E. Build two possible arrangements.
- Q6. Complete the series: 3, 8, 18, 38, 78, ?. Write the pattern in one sentence.
- Q7. Pick the sentence with the clearest business communication and rewrite the weakest option.
- Q8. Write a 90-word email asking HR to reschedule an interview politely and professionally.

Section C: Programming Logic and Coding

- Q9. Trace a loop that updates sum only when i is divisible by 3 or 5. State final output for n = 20.
- Q10. Given an array, return the second highest distinct value. Include edge cases for duplicates and negative values.
- Q11. Count frequency of each character in a string and print the first non-repeating character.
- Q12. Write pseudocode to reverse words in a sentence without reversing the characters inside each word.

Section D: SQL and CS Fundamentals

- Q13. Write a query to find departments with average salary above the company average.
- Q14. Explain primary key vs unique key vs foreign key with one example table.
- Q15. Explain process vs thread, normalization, indexes, HTTP status codes, and OOP inheritance with examples.
- Q16. What happens when you type a URL in browser? Explain DNS, TCP, HTTP, server, and response in simple words.

Section E: HR and Communication

- Q17. Tell me about yourself for a fresher role at service based companies. Keep the answer under 60 seconds.
- Q18. Why service based companies? Connect learning, role fit, adaptability, and your preparation evidence.
- Q19. Tell me about a failure. Use STAR format and end with what changed in your behavior.
- Q20. Explain your best project to a non-technical interviewer in 90 seconds.

Model Hints

- Q1 hint: if 60% takes 36 minutes, 100% takes $36 \times 100 / 60 = 60$ minutes.
- Q3 hint: net work per hour = $1/12 - 1/18 = 1/36$, so tank fills in 36 hours.
- Q10 hint: track largest and second largest distinct values; handle fewer than two distinct values.
- Q13 hint: use GROUP BY department and HAVING AVG(salary) > (SELECT AVG(salary) FROM employees).

Solution Checklist

- Write assumptions before solving aptitude.
- Show dry run for every coding answer.
- Use STAR format for HR answers.
- Keep final answer short, structured, and interview-friendly.
- After the set, classify mistakes as concept gap, formula gap, reading error, time trap, or panic.

Original OA and Interview Practice

- This set is designed for product based companies style product preparation: DSA depth, CS fundamentals, design thinking, project ownership, and behavioral clarity.

Section A: DSA Pattern Practice

- Q1. Return indices of two numbers adding to target. Explain hash map time and space complexity.
- Q2. Merge overlapping intervals and list the sorting invariant.
- Q3. Detect cycle in a linked list, then find the cycle start.
- Q4. Find shortest path in an unweighted graph and explain why BFS works.
- Q5. Find the longest substring without repeating characters. Explain sliding window movement.
- Q6. Search an element in a rotated sorted array. Explain the binary-search decision rule.
- Q7. Given daily temperatures, return days to wait for warmer temperature. Explain stack invariant.
- Q8. Count islands in a grid. Explain visited marking and boundary checks.

Section B: DP and Optimization

- Q9. Count ways to climb stairs with 1 or 2 steps, then optimize memory.
- Q10. Find maximum subarray sum and explain why local optimum works.
- Q11. Find coin change minimum coins. Define state, transition, base case, and impossible state.
- Q12. Given jobs with start, end, profit, select maximum profit non-overlapping jobs. Explain sorting + DP idea.

Section C: LLD and HLD

- Q13. LLD: Design a parking lot with classes, responsibilities, pricing, and edge cases.
- Q14. LLD: Design a rate limiter. Compare fixed window, sliding window, and token bucket.
- Q15. HLD: Design a tiny URL service. Cover APIs, storage, cache, collisions, and scaling.
- Q16. HLD: Design a notification system. Cover queues, retries, idempotency, and failure handling.

Section D: CS Core and Project Depth

- Q17. Explain database indexing and when an index can hurt performance.
- Q18. Explain process vs thread and how context switching affects performance.
- Q19. Explain HTTP status codes, cookies, JWT, and session tradeoffs.
- Q20. Describe a time you handled ambiguity while preparing for product based companies. Use a real example.

Model Hints

- Q1 hint: hash map stores complement or seen values; time $O(n)$, space $O(n)$.
- Q3 hint: Floyd cycle detection gives meeting point; reset one pointer to head to find cycle start.
- Q8 hint: DFS/BFS from every unvisited land cell; each cell is visited once.
- Q14 hint: mention data model, request key, time bucket, concurrency, and cleanup.

Solution Checklist

- State brute force first, then optimized approach.
- Confirm input constraints before coding.
- Dry run one normal case and one edge case.
- End with complexity and production tradeoffs.
- For design rounds, define requirements before drawing components.

PYQ-Style Attempt Protocol

- Step 1: identify topic and difficulty before solving.
- Step 2: write the first approach, even if it is brute force or long method.
- Step 3: optimize only after the basic idea is correct.
- Step 4: dry run one normal case and one edge case.
- Step 5: add answer, time, mistake type, and retest date to the log.

Pattern References

- Common placement patterns: percentage change, ratio split, time-work efficiency, seating arrangement, sentence correction, loop tracing, SQL grouping.

Answer Review Table

Mark	Meaning	Next Action
Solved	correct under timer	revise after 7 days
Guessed	answer came without method	solve two similar questions
Wrong	concept or execution gap	write mistake rule and reattempt tomorrow
Skipped	time or confidence issue	learn shortcut or move to lower difficulty

Execution Dashboard

Metric	Target	Review Frequency
Aptitude Sets	5 timed sets per week	every Sunday
CS Core Notes	4 topics per week	twice a week
Mock Tests	1-2 per week	same day analysis
Interview Practice	5 answers per week	record and review

Weekly Review Ritual

- Pick the top 5 mistakes, rewrite the correct rule, and solve one similar question.
- Remove one weak resume line or prepare one stronger project answer.
- Decide the next week from evidence, not mood.

Final Self Audit

Score	Meaning	Action
0-40	reading stage	finish basics and easy questions
41-70	practice stage	increase timed mocks and mistake review
71-85	interview-ready	polish project, HR, and weak topics
86-100	selection-ready	maintain revision and avoid overloading new material

Source Note

- This is original Study Hub premium material for learning and practice. It is not copied from official or proprietary company papers.